Introduction:

Climate change poses increasingly urgent challenges to societies, economies, and ecosystems worldwide. Understanding and predicting climate trends (especially temperature) is key for environmental planning, agriculture, energy systems, and disaster preparedness. I also found it fascinating that NOAA collects data from over 12,000 weather stations worldwide as of 2023 (almost five times more than in 1970).

This project explores historical climate data from NOAA's GSOD (Global Summary of the Day) to predict monthly average temperatures using machine learning. We leverage Apache Spark to process over 30GB (over 150 million records) of raw weather records from 1970–2023 and train models on decades of data to evaluate how well past patterns can predict future temperatures.

This approach demonstrates how scalable data pipelines and predictive models can be used to estimate temperature trends globally, potentially aiding in regional forecasting and policy analysis.

Some Figures from Data Exploration and Model:

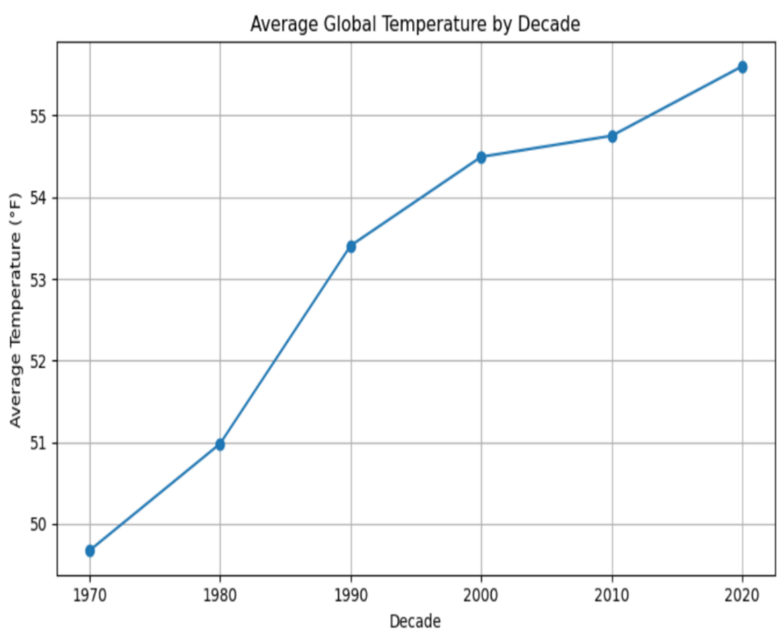
```
Total records: 151,898,241
+----+-------+
|year|  count|
+----+-------+
|2023|4038747|
|2022|3994717|
|2021|4026550|
|2020|4119205|
|2019|4166563|
|2018|4130871|
|2017|4156593|
|2016|3997629|
|2015|4021868|
|2014|3930772|
+----+-------+
```

Amount of records in the data set is over 150 million and currently averaging approximately 4 million per year globally.

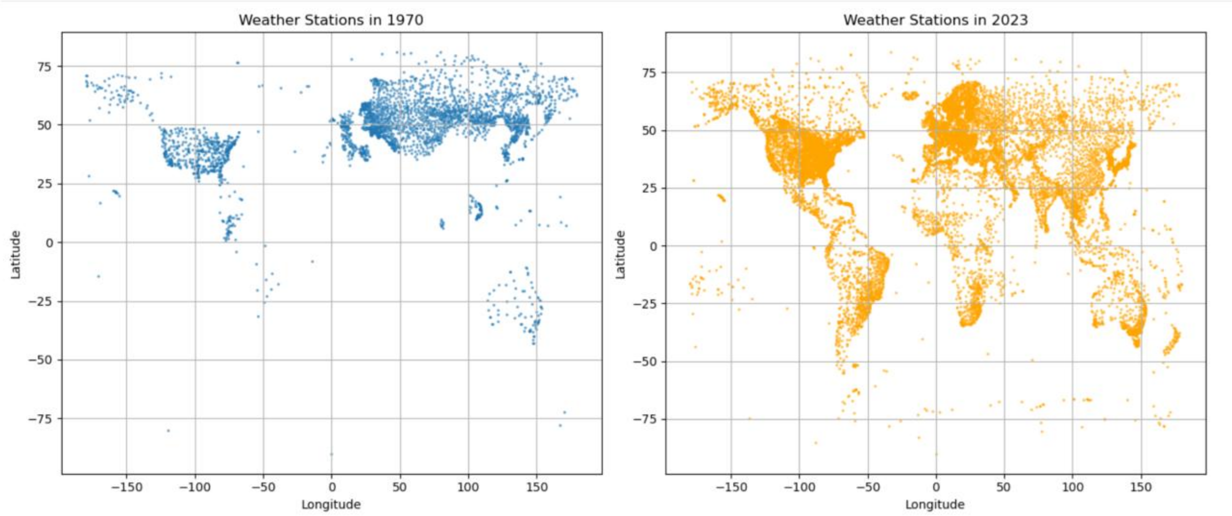
```
 Bottom 3 Coldest Countries (2011-2020):
+----+--------+
|CTRY|avg_temp|
+----+--------+
|  AY|   -4.16|
|  AQ|   12.72|
|  GL|   23.37|
+----+--------+
only showing top 3 rows

 Top 3 Warmest Countries (2011-2020):
+----+--------+
|CTRY|avg_temp|
+----+--------+
|  DJ|   89.02|
|  CD|    87.1|
|  NG|   84.97|
+----+--------+
```

The coldest and warmest countries from 2011-2020



Average global temperature increasing over the decades from 1970 to 1920



This plot I found extremely interesting as each dot represents a weather station that NOAA pulls daily data from comparing 1970 to 2023. Just looking at the dots in 2023, you can clearly see the outline of all continues including Antarctica.

```
RMSE Results:
  Train RMSE : 3.71
  Test RMSE  : 3.82
  Test R²    : 0.97
  Test MAE   : 2.76
```

2011-2020 Monthly Average Temperature: Actual vs. Predicted

month_name	actual_avg_temp	predicted_avg_temp
Jan	39.49	39.45
Feb	41.95	41.81
Mar	48.63	48.22
Apr	56.31	55.82
May	63.16	62.55
Jun	68.67	68.45
Jul	71.63	71.11
Aug	70.89	70.51
Sep	65.57	65.31
Oct	57.61	57.45
Nov	48.85	48.76
Dec	42.27	42.19

Here I attempted to predict the global monthly average temperature of the 2011-2020 decade based on historical weather data.

Methods

Downloading of Data – The first step was to download the millions of records from NOAA's website. The attached notebook shows how this was done but in essence I created a script to download each weather stations csv files and separate them by year. I then converted the data to parquet format for easier analysis.

Data Exploration – A typical way to begin data exploration is identifying the names of the columns and what data we're actually looking at. The GSOD has numerous columns with temperature, precipitation, snow depth, latitude/longitude, wind speed, among several others.

Something else important is identifying how many records the data encompasses which from 1970 – 2023 contained 151,898,241 and the count of weather stations increased from 2,749 in 1970 to 12,217 in 2023.

Preprocessing – Fortunately the GSOD database is well maintained and did not require much preprocessing. There were cases of placeholder values, air pressure, and windspeed being represented by very high values (i.e. windspeed of 999.9). I dropped rows containing these to prevent inaccurate results. I also removed rows with null values from temperature, latitude/longitude, elevation along with the three proceeding features as these were critical to my chosen model.

Model – The goal for my model was to try to predict the average monthly temperature of the 2011-2020 decade based on historical weather data. I selected the features mentioned in the preprocessing step (latitude, longitude, elevation, precipitation, air pressure, and windspeed. I then only selected weather stations with over 45 years of data to ensure reliable results and then I split the data into the training set of <2010 and the test set from 2011 to 2020.

I used Pyspark's GBTRegessor (Gradient Boosted Tree) with the feature vectors again being latitude, longitude, elevation, precipitation, air pressure, and windspeed with the target vector as average month temperature. A gradient boosted tree model builds multiple decision trees sequentially with each iteration attempting to correct errors in the previous tree. I felt this was a good choice trying to model something complex and non linear such as temperature. I felt 100 iterations with a map tree depth of 5 was a good balance to ensure accuracy and also avoid overfitting. Lastly, I set the subsampling rate to 0.8 or 80% to further reduce overfitting and improve generalization

Results:

RMSE Results:

Train RMSE: 3.71

Test RMSE: 3.82

Test R^2: 0.97

Test MAE: 2.76

2011–2020 Monthly Average Temperature: Actual vs. Predicted

Jan	39.49	39.45
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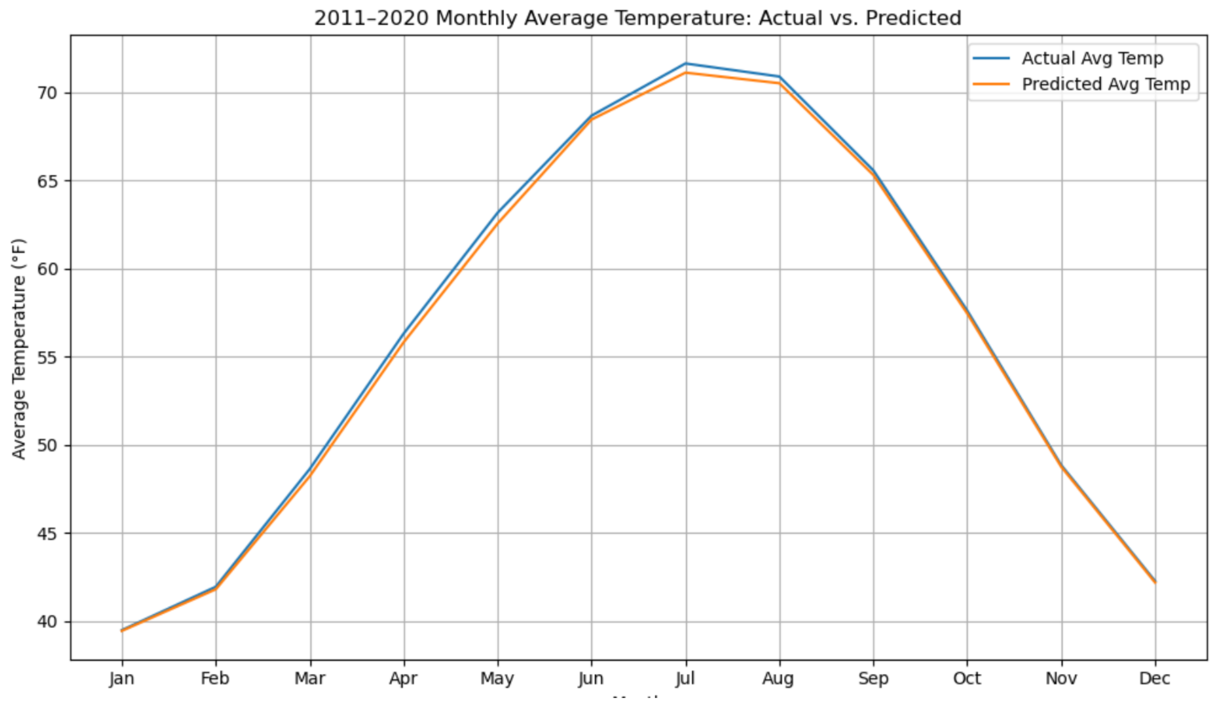

Discussion:

I started this project curious whether past weather patterns could actually help predict future temperatures. With climate being so complex, I didn't expect perfect results—but I wanted to see if a machine learning model could at least pick up on the broader patterns. I went with a Gradient Boosted Tree model because it's great at handling messy, nonlinear data like this without needing a lot of manual setup. I made sure to only include stations with at least 45 years of data to keep things consistent over time.

The model ended up performing really well: Train RMSE was 3.71°F, Test RMSE was 3.82°F, and the Test R^2 score came out to 0.97, which means it explained almost all the variation in the data. The Test MAE was 2.76°F, showing that most predictions were within a couple degrees of the actual temperature. When I looked at the monthly results from 2011–2020, the predictions stayed impressively close to the real values. For example, July had an actual average of 71.63°F and the model predicted 71.11°F. That kind of accuracy is pretty encouraging. That said, I did notice it underestimated summer temps slightly, which might mean the model is smoothing out the more extreme highs. Overall, though, it captured seasonality and geographic patterns well and felt reliable within the time frame I tested.

Conclusion

Looking back, I'm happy with how this project turned out— especially being able to take millions of raw climate data entries spanning more than 50 years and turn it into something visual and then also building a model that could make decent predictions. One thing I'd like to do with this in the future is explore is how the model behaves at a more regional level.

Predicting global averages is useful, but it would've been interesting to zoom in on specific areas or climate zones and see how well the model generalizes there and if there are certain regions affecting average temperatures more than others.

Another thing I'd like to do is build a model I could actually try predicting beyond 2020. That said, building this full pipeline—from downloading hundreds of thousands of files, to cleaning and modeling, and then evaluating—taught me much about working with Spark and machine learning. There's definitely much more to do, but this project gave me a solid foundation.

Statement of Collaboration

I completed this project myself (largely due to work hours) from beginning to end. I did not want to hinder/not be available for team communications.